



Prodromal Parkinson disease subtypes — key to understanding heterogeneity

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Abstract | In Parkinson disease (PD), pathological processes and neurodegeneration begin long before the cardinal motor symptoms develop and enable clinical diagnosis. In this prodromal phase, risk and prodromal markers can be used to identify individuals who are likely to develop PD, as in the recently updated International Parkinson and Movement Disorders Society research criteria for prodromal PD. However, increasing evidence suggests that clinical and prodromal PD are heterogeneous, and can be classified into subtypes with different clinical manifestations, pathomechanisms and patterns of spatial and temporal progression in the CNS and PNS. Genetic, pathological and imaging markers, as well as motor and non-motor symptoms, might define prodromal subtypes of PD. Moreover, concomitant pathology or other factors, including amyloid- β and tau pathology, age and environmental factors, can cause variability in prodromal PD. Patients with REM sleep behaviour disorder (RBD) exhibit distinct patterns of α -synuclein pathology propagation and might indicate a body-first subtype rather than a brain-first subtype. Identification of prodromal PD subtypes and a full understanding of variability at this stage of the disease is crucial for early and accurate diagnosis and for targeting of neuroprotective interventions to ensure efficacy.

Idiopathic Parkinson disease (PD) has many emerging facets of temporal and spatial progression within the CNS and PNS. On the basis of pathological findings and clinical observations of a slowly progressing neurodegenerative process, several phases of PD have been defined¹. Before the onset of neurodegeneration, there is a risk phase to which genetic and environmental factors contribute and in which substantia nigra hyperechogenicity can be observed². The subsequent preclinical phase is characterized by the initiation of progressive neurodegenerative pathology before any clinical symptoms or signs are evident. No markers have yet been established to reliably identify individuals at this stage of the disease.

The later prodromal phase is defined by the emergence of observable signs or symptoms of neurodegeneration³. Markers of the prodromal phase include REM sleep behaviour disorder (RBD), olfactory loss, autonomic dysfunction, depression (with or without comorbid anxiety), mild motor signs, and pathological imaging markers of the presynaptic dopaminergic system and the cardiac sympathetic system. These markers have predictive value for clinical PD, although their specificity varies considerably. Several lines of

compelling evidence indicate that the prodromal phase precedes clinical diagnosis of PD by 10–20 years^{4–7}. Fully manifest bradykinesia with rest tremor and/or rigidity⁸ define the transition to the clinical motor phase of PD, according to International Parkinson and Movement Disorder Society (MDS) criteria. Pathologically, this phase is characterized by nigrostriatal synucleinopathy and advanced neurodegeneration.

Within this broad characterization of PD phases, the onset and progression of motor and non-motor symptoms can differ enormously between individuals, not only in the clinical phase but also in the prodromal phase⁹. Some features, such as RBD, early cognitive deficits, autonomic dysfunction and genetic risk factors, are associated with faster progression^{10,11} and a shorter life expectancy^{12,13}. In addition, several biological processes that are inherent to normal ageing, such as genomic instability, telomere attrition, epigenetic alterations, loss of proteostasis, mitochondrial dysfunction, cellular senescence, deregulated nutrient sensing, stem cell exhaustion and altered intercellular communication, are also likely to be relevant to the progression of PD¹⁴. The initiation and progression of PD pathology can also be heterogeneous between patients, and the extent and

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Key points

- Heterogeneity of Parkinson disease (PD) starts in the prodromal phase.
- Pathological spread, imaging markers and the onset and progression of motor and non-motor symptoms are variable in prodromal PD.
- The variability in clinical phenotype suggests that subtypes of prodromal PD can be defined.
- Possible subtypes of prodromal PD are REM sleep behaviour disorder subtypes, brain-first and body-first subtypes, genetic subtypes and biological subtypes.
- Variability in prodromal PD must result from different pathophysiological mechanisms; defining prodromal subtypes is the first step to identifying the biological difference.

distribution of pathology at the prodromal to clinical stage can vary. Furthermore, different organs^{15,16}, various cellular and systemic processes, biofluid-borne pathogens and risk factors can be involved in the aetiology and spatial progression of PD¹⁷.

The borders of any complex degenerative disease can never be defined perfectly, and this is particularly challenging for PD. If a conceptual framework is applied in which a disease has established aetiologies, syndromes are organized collections of symptoms with no established aetiology, pathology is a marker of a disease and the pathophysiology and aetiology relate to the cause, the position of PD is unclear. Although PD has no defined, unified pathophysiology, much is known about its aetiology, so it could be described as a disease. However, PD is heterogeneous so could also be considered a syndrome¹⁸, or as an umbrella term under which different and overlapping clinical and pathological subtypes fit. Moreover, given that the pathology is similar in dementia with Lewy bodies (DLB) and pure autonomic failure, suggesting a common and overlapping pathophysiology, PD could be considered a subclassification under an even larger umbrella of Lewy body disorders or neuronal synucleinopathies^{3,19–22}.

Given that clinical PD is simultaneously heterogeneous and can be classified under a larger umbrella, the same is presumably true for prodromal PD. Therefore, understanding variability in the prodromal phase of PD is key to a holistic understanding of the full spectrum of PD and could pave the way for early neuroprotective therapies that are specific to prodromal subtypes and consequently have high efficacy. In this Review, we provide an overview of the current clinical and pathophysiological understanding of prodromal PD before discussing evidence of variable pathological propagation in PD and how this variability could relate to prodromal subtypes.

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Prodromal PD — definition and diagnosis

By the time PD can be diagnosed on the basis of its cardinal motor symptoms, the neurodegeneration — not only in the substantia nigra pars compacta but more widely — has been advancing for many years^{23,24}. Evidence from prospective studies not only continues to confirm the concept of prodromal PD but has enabled the development of PD prediction models^{25–27}. Risk markers, including genetic factors^{28–30}, environmental and lifestyle factors and concomitant diseases³¹, have consistently been associated with an increased risk of PD. In addition, biomarkers such as low plasma levels of urate and hyperechogenicity of the substantia nigra (when performed by an experienced investigator, as accuracy depends on experience)² can be combined with other markers to help identify high-risk states^{26,32}. Prodromal markers in the MDS research criteria for prodromal PD include subtle motor signs^{11,33}, a diverse list of non-motor symptoms^{5,7,25,26} and pathological findings upon dopaminergic functional imaging³⁴. For some markers, a clear distinction between their status as a risk marker or prodromal marker cannot always be made. Moreover, some are risk markers and prodromal markers at different times. For instance, depression can be a PD risk marker in early life but can also indicate neurodegeneration in the prodromal phase⁶.

The evidence for several of these established risk and prodromal markers of PD has recently been reviewed²⁶, and these markers form the basis of the MDS research criteria for prodromal PD^{25,26}. Inclusion of a given marker in the MDS prodromal criteria generally required that the marker had predicted PD onset in at least two independent prospective studies. With the use of a naive Bayesian classifier, these criteria enable practical and evidence-based calculation of the probability that an individual has prodromal PD on the basis of their age and marker profile (Supplementary Box 1 and Supplementary data).

The original research criteria²⁵ have been validated in different cohorts, and have relatively high specificity and a high positive predictive value for the conversion from probable prodromal PD (>80% probability) to clinical PD^{35–37}. The sensitivity of the criteria varies and depends primarily on the depth of marker assessment (highly predictive markers are required for high sensitivity) and on the time to PD diagnosis (sensitivity drops at early time points, when many features have not had time to develop)³⁸. The criteria have been updated once and will continue to be updated regularly²⁶. In the latest version, the likelihood ratio estimates of markers were updated and four new markers were added. In addition, a new approach for considering genetic risk markers was developed, which incorporated age-related penetrance of PD (for *GBA*²⁸ and *LRKK2* (REF.²⁹) mutations) and polygenic risk scores calculated from low penetrance variants³⁰.

Compelling evidence supports the inclusion of several other promising markers but fewer than two prospective studies had been completed for these when the prodromal criteria were updated. For instance, a meta-analysis has shown that detection of phosphorylated α -synuclein in the skin had a pooled sensitivity

of 76% and a specificity of 100% for identification of patients with PD, and detection in the submandibular gland had a sensitivity of 66% and a specificity of 96%³⁹. Cross-sectional studies have shown that this α -synuclein pathology is present in the majority of people with RBD — given that >80% of people with RBD develop PD or DLB and α -synuclein pathology is more common in RBD-positive PD than in RBD-negative PD, α -synuclein in the skin or submandibular gland is likely to be a highly specific marker of prodromal PD⁴⁰. In addition, neuroimaging modalities, such as PET, single-photon emission CT (SPECT) and MRI, have high potential for measurement of sensitive and specific markers of progressive neurodegeneration in prodromal PD^{15,41}. Similarly, subtle abnormalities and progressive changes in motor^{42,43} and non-motor markers^{44,45} in prodromal PD can be detected with wearable and smartphone-based sensor technologies. In these areas, the results of ongoing prospective studies are awaited.

Despite this progress in identification of prodromal PD, several missing pieces remain. For instance, the predictive values of risk and prodromal markers can depend on the time to PD diagnosis, marker-specific lead times and increases in the strength of the marker in the time that precedes PD diagnosis. If markers are correlated or have pathophysiological interactions, their predictive values might be higher or lower than if they are considered independently^{26,46}. Furthermore, the clinical trajectories of people with RBD, who are likely to be a fairly homogeneous prodromal subgroup, have been investigated more extensively^{7,47} than those of population-based cohorts, even though the findings in these cohorts would be more generalizable to the heterogeneous spectrum of (prodromal) PD.

Another important consideration is that prodromal PD is considered to be a bona fide stage of PD. Studies of prodromal markers have so far considered the end point of phenoconversion to clinical PD as a zero–one

variable; this is an oversimplification that might be misleading. For example, analysis of sample sizes for clinical trials in RBD suggests that milestones of decline on quantitative measures (for example, motor and/or cognitive scales) are more efficient outcomes than phenoconversion (and are also less difficult to define⁴⁸). The zero–one distinction might also be an oversimplification with respect to disease heterogeneity, and prodromal PD subgroups might be needed. From the perspective of patients, prodromal PD is also highly relevant because patients often have considerable disease burden from a variety of (especially non-motor) symptoms long before they are ‘labelled’ and treated as patients with PD⁴⁹.

PD heterogeneity and the prodrome

Before exploring the potential heterogeneity of prodromal PD, it is useful to review what is known about the heterogeneity of clinical PD. Several different phenotypes of PD have been identified, each with different clinical characteristics and progression (FIG. 1). Broadly speaking, PD subtypes proposed to date have been based on either single features or multidomain phenotypes.

Single-feature classifiers

Age. Manifestations of PD differ considerably according to age — the most striking differences are seen in motor fluctuations and cognition. In general, younger patients are more likely to experience a dramatic motor benefit with levodopa treatment and subsequently more motor fluctuations and dyskinesia. By contrast, dementia occurs almost exclusively in older patients, and cognitive impairment is more related to current age than to disease duration⁵⁰. The mechanism by which age affects phenotype is unclear; plausible mechanisms include a higher prevalence of co-morbid pathology (which is common in autopsy samples from people with PD)^{51,52} and a more diffuse vulnerability to the spread of α -synuclein pathology or α -synuclein-associated damage. Numerous

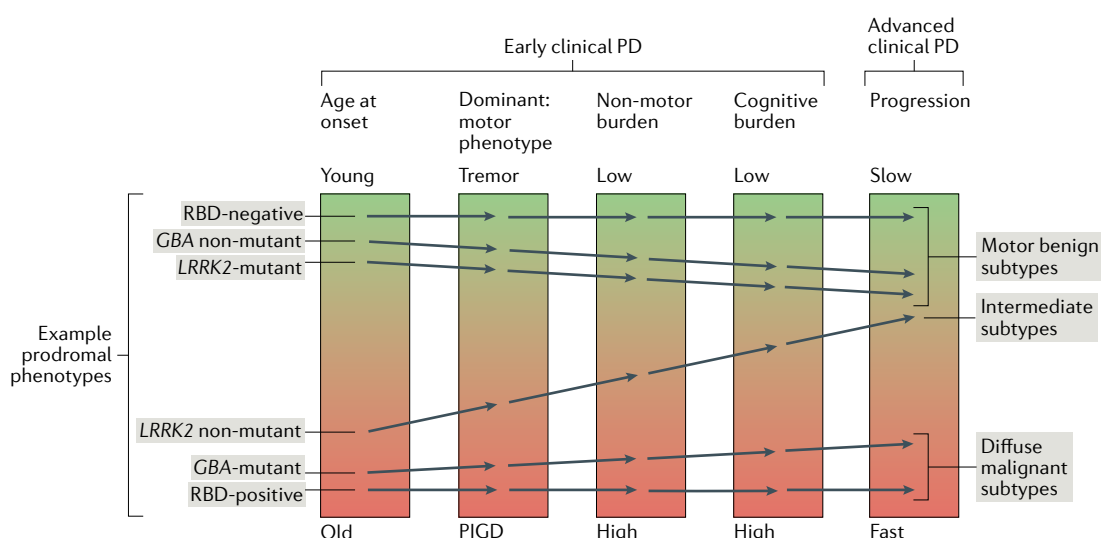


Fig. 1 | Characteristics of prodromal PD subtypes. Each trajectory shows the characteristics of a different prodromal subtype in early clinical Parkinson disease (PD) and the associated rate of progression in advanced clinical PD. For each subtype, the identifying clinical markers differ at the earliest stages of disease and could affect the stage at which a diagnosis of PD is made. PIGD, postural instability and gait disturbance; RBD, REM sleep behaviour disorder.

exceptions to this rule exist; many 90-year-old patients experience dyskinesia, and dementia can develop before the age of 60 years, particularly in some patients with genetic mutations, such as a mutation in the *GBA* gene or *SNCA* gene triplication. Moreover, age is a continuous variable and no clear cut-off can be defined to distinguish ‘young PD’ and ‘old PD’, which limits the use of age as a basis for categorical subtypes.

Motor phenotypes. Patients with tremor in PD often do better on average than those without⁵³. Several classifications have been used to distinguish PD with tremor from ‘other’ types of PD. Most approaches reveal prognostic differences, although these differences are relatively small. Gait, rather than tremor itself, could be the primary driver of differences between motor phenotypes; postural-instability-gait-predominant phenotypes are associated with faster cognitive decline and a higher prevalence of many non-motor symptoms than other forms. The observed links between motor phenotype and prognosis are partially confounded by the promptness of diagnosis, because tremor is an effective early diagnostic signal of PD, and by diagnostic errors (severe atypical parkinsonian conditions, such as multiple system atrophy (MSA) and progressive supranuclear palsy, do not often present with tremor)³. Exceptions to the finding that gait subtypes are associated with worse PD also exist; for example, patients with *LRRK2* mutations classically have prominent gait dysfunction but an otherwise good prognosis, especially with respect to non-motor features, such as dementia, a clinical history of RBD and autonomic dysfunction⁵⁴.

Dementia at onset. Previously, dementia at onset was considered to be an exclusion criterion for PD. However, according to the current MDS criteria, PD can now be diagnosed irrespective of the presence or onset time of dementia^{3,22}. Therefore, PD with early-onset dementia could be considered a subtype. The phenotype of patients who are diagnosed with DLB generally resembles the so-called diffuse-malignant subtype of PD (see Data-driven clustering), including a limited response to levodopa, prominent gait abnormalities, a clinical history of RBD and orthostatic hypotension. This observation is consistent with the fact that people with idiopathic RBD (iRBD) have an approximately equal risk of developing PD or DLB first and, therefore, are often included in this same diffuse-malignant subtype.

Co-morbid pathology could be an important driver of dementia-onset PD — one autopsy study indicated that 89% of patients with DLB had co-morbid amyloid- β and α -synuclein pathology, and the same was seen in 38% of patients with PD dementia^{51,52}. If patients with DLB are included under a Lewy body or PD umbrella, then diffuse-malignant PD subtypes could be much more common than apparent when only patients who develop parkinsonism first are considered. Criteria for prodromal DLB were published in 2020 (REF.⁵⁵); the purpose of these criteria was limited to identification of patients with mild cognitive impairment with underlying Lewy body pathology. As the fields of PD and DLB continue to converge, unified criteria for prodromal PD

and DLB (which together can be classed as Lewy body disease) must be developed.

Multi-domain phenotyping

Data-driven clustering. Advanced statistical, data-driven techniques, such as machine learning, can be used to explore variations between patients who have been diagnosed with a single disease. The most commonly used approach is unsupervised clustering, in which, in simple terms, a group of patients are phenotyped, phenotypic variables are standardized and the clustering algorithm sorts patients into subtypes on the basis of these variables. This type of cluster analysis has the major advantages of being hypothesis-free, enabling discovery of otherwise unanticipated variables, and capable of simultaneous interrogation of many motor and non-motor clinical features⁵³. As knowledge of non-dopaminergic and non-motor features has increased, non-motor variables have increasingly been included in such studies, which has dramatically improved multidomain clustering. From these studies, cognitive impairment, sleep disorders, autonomic dysfunction and hallucinations have emerged as important determinants of subtypes. When non-motor and motor variables are included in these models, many non-motor variables are stronger determinants of subtype than are motor features^{11,53}.

One major disadvantage of data-driven clustering is that study-specific factors, such as patient recruitment bias, the variables included and the statistical methods used, mean that each cluster solution is its own entity, making it difficult to identify unifying themes between studies. Regardless, perhaps the most consistent group that emerges from these different analyses is a subtype with a poor prognosis, also described as a diffuse-malignant subtype^{11,12,53}, where ‘diffuse’ implies a broad pattern of non-motor and motor features and ‘malignant’ implies a poor overall prognosis. This subtype is broadly characterized by an older age at onset, dominant non-motor manifestations, especially in the cognitive domain, and more rapid progression to milestones such as dementia, a requirement for gait assistance, institutionalization and death^{11,12,53,56}. Patients with the diffuse-malignant subtype experience a high burden of non-motor features from the earliest stages, particularly RBD, dysautonomia and cognitive impairment. Many patients with idiopathic RBD develop this diffuse-malignant subtype of PD when they phenocconvert. This subtype also overlaps with DLB and probably includes many people who have been diagnosed with DLB. Many cluster analyses have also identified a subtype at the other end of the spectrum characterized by a younger age at onset (but not always), less non-motor involvement, more tremor predominance (although this is not invariable), slower progression and a better prognosis with respect to both cognitive and motor function, although motor fluctuations often occur^{10,53–55}.

When considering the findings of any cluster analyses, we must also take into account findings from studies in people with de novo PD because instability of motor phenotypes and transition from one phenotype to another have been reported in de novo PD^{57,58} and moderate stages of the disease⁵⁹. These observations challenge

the definitions of stable motor subtypes. Associations between the subtypes and biomarkers and clinical markers are generally modest. This difficulty could be a reflection of the lack of biomarkers for PD as much as a reflection of the clinical phenotyping. Nevertheless, some associations are seen. Cerebrospinal fluid (CSF) analysis has shown that patients with diffuse-malignant subtypes of PD are more likely to have an Alzheimer-like CSF profile, with high levels of amyloid- β and low levels of tau⁶⁰. Dopamine transporter (DaT) SPECT findings might enable identification of the more severe clinical phenotypes — caudate dopaminergic denervation is a possible predictor of subtypes that involve extensive cognitive impairment^{60,61}. MRI might be useful for seeing atrophy and disruption of connectivity in patients with diffuse-malignant subtypes, although studies remain limited^{62,63}. Finally, autopsy and biopsy studies have shown that patients with RBD or diffuse-malignant PD have a higher burden of α -synuclein deposition across the brain and in the gut than people with other subtypes, suggesting that the pathophysiology in these patients is driven by α -synuclein to a greater extent than that of other subtypes^{12,40,64}.

Non-motor endophenotypes. A different approach to exploring heterogeneity is to classify clinical symptoms rather than patients. Use of this approach has led to the proposal of non-motor endophenotypes that fit into neurotransmitter-based models⁶⁵ in which neurochemical deficits in adrenergic, serotonergic, opiodergic and cholinergic pathways occur in addition to dopaminergic involvement^{66,67}. In one cluster analysis of non-motor symptoms, various phenotypic traits were united into symptom clusters, including a depression and excessive sleepiness cluster and a cognitive impairment and apathy cluster. For some of these endophenotyping solutions, the presumed underlying pathophysiology includes distinct routes of spread of pathology, resulting in different dominant patterns of involvement in PD. For example, RBD, daytime somnolence and dysautonomia are the prominent non-motor manifestations in patients with a brainstem dominance of pathology. By contrast, depression, fatigue and early anosmia characterize an olfactory-to-limbic pattern of pathology and early cognitive deficits and apathy are features of a neocortical route of spread. These three patterns are the dominant non-motor endophenotypes^{66,67}.

Biomarker-based subtypes. Few attempts to define PD subtypes on the basis of biomarkers have been successful. One obvious obstacle is the fact that PD is defined clinically; therefore, omitting clinical manifestations from subtyping solutions is problematic. As a result, rather than being used to define distinct subtypes themselves, biochemical markers have largely been assessed for their association with clinical features or with previously defined clinical subtypes of PD. The biomarkers studied have largely been CSF markers and α -synuclein deposits in tissue biopsies.

CSF biomarkers in PD are less robust than those in, for example, Alzheimer disease (AD), so subtyping of PD based on such markers remains difficult. One study

of patients with de novo PD showed that CSF levels of α -synuclein are low in the non-tremor-dominant subtype of PD⁶⁸ and that CSF levels of amyloid- β and phosphorylated tau are low in the postural instability and gait disturbance subtype⁶⁹. In patients with PD, an AD-like CSF profile — that is, low levels of amyloid- β and high levels of tau — correlates with older age, presence of the APOE ϵ 4 allele, early cognitive impairment and reduced grey matter in the frontal cortex⁷⁰. The diffuse-malignant subtype of PD has also been associated with this AD-like CSF profile⁶¹. These observations suggest that an underlying co-morbid AD pathology is present in some patients with this subtype of PD and can substantially influence the clinical presentation. Patients with classically-defined DLB also have low CSF levels of amyloid- β , again suggesting overlap with diffuse-malignant PD subtypes.

Protein amplification assays, which were developed in the prion disease field, have shown great promise for accurate CSF-based diagnosis of PD⁷¹. These assays can detect α -synuclein seeding activity across the spectrum of Lewy body disorders including PD, iRBD, DLB and pure autonomic failure with a sensitivity and specificity of >95%⁷². Although this finding suggests a core, homogeneous cause of PD, detailed characterization of α -synuclein might reveal differences in α -synuclein conformations (so-called strains) between different Lewy body disorders and between subtypes of PD.

Skin biopsies have revealed deposits of α -synuclein in dermal nerve fibres in various synucleinopathies, including PD, DLB and MSA. Two studies have shown that this abnormal deposition is detectable during the prodromal stage in individuals with RBD^{16,73}. As a minimally invasive method, skin biopsy seems to be a suitable candidate measure to screen for prodromal PD during enrolment in trials of disease-modifying, α -synuclein-based therapies. α -Synuclein deposits can also be detected in biopsy samples of other peripheral tissues, such as submandibular or minor salivary glands and submucosal enteral tissue — these biopsies are more invasive than skin biopsy but could be considered as alternatives^{74–76}. Whether the characteristics of PD differ between patients with positive biopsies and those without remains to be seen.

Several studies have identified associations between other biochemical markers, such as amyloid- β , apolipoprotein A1, C-reactive protein, uric acid and vitamin D in the blood or CSF, and different subtypes of PD. However, these studies involved post hoc comparisons of biomarkers between clinically defined PD subtypes, so the markers identified do not enable a new method of categorizing PD^{77,78}.

Imaging prodromal PD

Given that parkinsonism emerges when >50% of striatal dopamine terminals are lost⁷⁹, dopamine imaging is well suited to the monitoring of preclinical loss in this system. Indeed, presynaptic dopamine imaging markers are abnormal in approximately 30–40% of people with idiopathic RBD⁸⁰, although abnormalities are more extensive among patients with newly diagnosed PD⁸¹. Several studies have shown that pathological DaT SPECT scans

indicate imminent phenoconversion in people with iRBD or hypsmia^{34,82,83}.

Far less dopamine imaging data are available on other types of prodromal PD. In studies of people with non-manifest mutations associated with PD, only 11% of people with *LRRK2* mutations and 3% of people with *GBA* mutations exhibited a deficit upon DaT imaging⁸⁴. The contrast with findings in patients with iRBD are not surprising, as RBD symptoms signify that brainstem neurodegeneration is already pronounced, whereas imaging markers are likely to be normal in people with mutations but no symptoms.

Advanced MRI sequences can also be used to visualize the substantia nigra itself. This approach has demonstrated that dorsal nigral hyperintensity is reduced in 27% of patients with idiopathic RBD⁸⁵. Concordantly, transcranial sonography has revealed nigral hyperechogenicity in 35–50% of patients with iRBD, although the same was seen in >10% of healthy controls^{86,87}.

A small number of imaging studies have been conducted to investigate the involvement of other CNS nuclei and neurotransmitter systems in prodromal PD. Imaging of neuromelanin has revealed a low signal in the locus coeruleus in 40–50% of people with iRBD, whereas this signal is normal in people with de novo PD and no RBD^{15,88}. This finding is consistent with the notion that RBD is caused in part by damage to the coeruleus and subcoeruleus nuclei. Imaging with ¹¹C-MeNER PET has shown that noradrenergic transporter density is low in the sensory–motor cortices of people with iRBD, which could signify an early mechanism to compensate for incipient dopaminergic neurodegeneration⁸⁹. Upregulation of vesicular cholinergic transporters in the brainstem has also been observed in people with iRBD, which could signify another compensatory mechanism⁹⁰. Similarly, people with non-manifest *LRRK2* mutations exhibit regional upregulation of subcortical serotonin transporters and greater cortical cholinergic activity than controls without mutations^{91,92}. These findings demonstrate that some imaging measures of neurodegeneration can be complicated by homeostatic regulatory mechanisms, particularly during the early disease stages, thereby impeding their use as reliable markers of prodromal PD or disease progression.

Imaging can also be used to detect involvement of the peripheral autonomic nervous system, which is unique to Lewy body disorders among the neurodegenerative diseases. ¹²³I-MIBG scintigraphy has demonstrated that marked cardiac sympathetic denervation is nearly ubiquitous among patients with iRBD^{15,93,94}. By contrast, MIBG scans are normal in 40–50% of patients with de novo PD (at Hoehn and Yahr stage I)^{94,95}. These findings suggest that patients with prodromal PD but without RBD mostly have intact cardiac innervation⁹⁶. Studies of the parasympathetic innervation of the colon with ¹¹C-donepezil PET and CT produced similar findings: in patients with de novo PD and RBD and patients with iRBD the colonic PET signal is markedly reduced compared with that in healthy controls, whereas findings in patients with de novo PD but no RBD are similar to those in controls⁹⁶.

Overall, measures of presynaptic dopaminergic degeneration and of cardiac sympathetic denervation must currently be considered the most robust imaging markers of neurodegeneration in prodromal PD. The excellent signal-to-noise properties of these methods enables sensitive detection of subclinical neurodegeneration at the individual patient level. These measures can, therefore, assist in determining whether particular PD subtypes are characterized by initial degeneration of CNS structures or by early damage to the peripheral autonomic nervous system.

Possible subtypes of prodromal PD

A good deal of research has been conducted on the subtyping of clinical PD but not on the subtyping of prodromal stages. Below, we describe some ways in which prodromal PD could be subtyped, with direct implications for clinical PD. These categorical boundaries should not be considered absolute — some patients who seem to have one prodromal subtype could ultimately resemble patients with another subtype once clinical PD develops. In addition, these subtypes reflect our current knowledge; other ways to subtype prodromal PD are certain to exist but are yet to be discovered and developed.

REM sleep behaviour disorder subtypes

iRBD is by far the strongest clinical marker of prodromal synucleinopathy. Multicentre studies have shown that >80% of patients with iRBD will develop either PD, DLB or MSA (in rare cases), and studies with the longest follow-up periods suggest that nearly all will eventually develop a neurodegenerative disease^{97–99}. The phenoconversion rate is ~6–7% per year, and a slightly greater proportion of patients with iRBD progress to a parkinsonism-first phenotype than a dementia-first phenotype (55% versus 45%)⁴⁶. The risk can be stratified further with inclusion of other prodromal markers; for example, co-existent olfactory loss, subtle parkinsonism and mild cognitive impairment are associated with a hazard ratio of ~3 for phenoconversion compared with patients with RBD but without these markers, and autonomic markers, severe loss of REM atonia and DaT SPECT abnormalities are associated with a 1.5-fold to 2.5-fold increase in conversion rate⁴⁶. Numerous other biomarkers (for example, MRI, and electroencephalogram and electrocardiogram abnormalities) are also being explored as predictive markers of phenoconversion in RBD (see Imaging prodromal PD).

The study of patients with RBD can also be used to map progression of prodromal disease over time. For example, a 10-year retrospective analysis of people who phenoconverted from RBD to PD/DLB found that olfactory loss and autonomic dysfunction were the first markers to emerge at 15–20 years before phenoconversion, followed by motor and cognitive changes 4–8 years before phenoconversion⁷.

This knowledge of progression from RBD to Lewy body disease and its predictability means that people with iRBD are the ideal candidates for trials of neuroprotective treatments, as treatment can be provided to these people before irreversible nigral neurodegeneration has

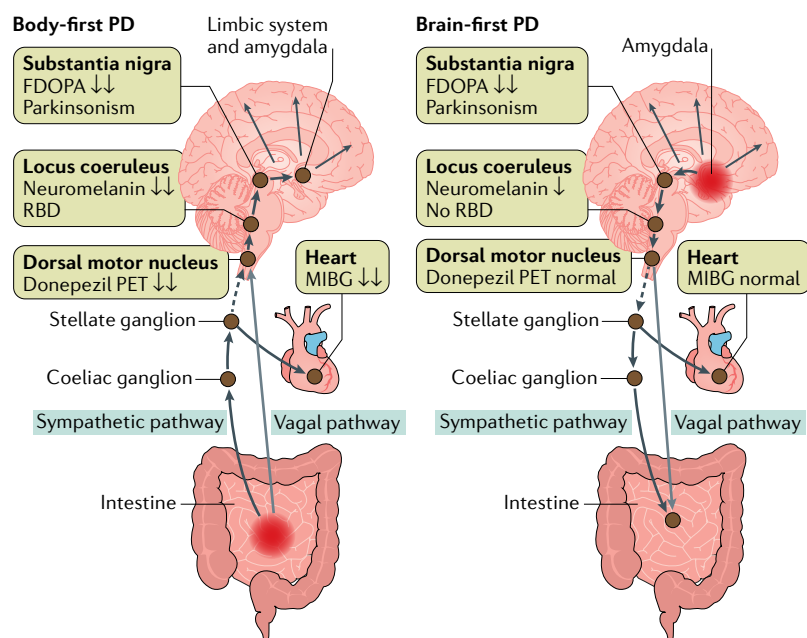


Fig. 2 | Characteristics of proposed body-first and brain-first subtypes of PD. Arrows indicate the direction of spread of α -synuclein pathology and the orange boxes indicate the predicted imaging findings and symptoms at the time when motor symptoms first emerge. In body-first Parkinson disease (PD; left), the first α -synuclein aggregation occurs in the gut and propagates through the autonomic nervous system to the spinal cord and brainstem. Autonomic dysfunction and REM sleep behaviour disorder (RBD) are prodromal features. In brain-first PD (right), the initial α -synuclein pathology arises in vulnerable structures of the brain, such as the limbic system, or enters via an olfactory route. The pathology then descends through the brainstem and RBD and dysautonomia generally develop after the onset of parkinsonism. During the later disease stages, nearly all patients have widespread pathology in these systems and their symptomatology therefore converges. FDOPA, fluorodopa. Adapted with permission from REF.⁹⁶, Oxford University Press.

occurred and at a time when symptomatic treatments for parkinsonism or dementia do not confound outcomes. Given the fact that patients with iRBD can develop parkinsonism and dementia, treatments that target general mechanisms, such as α -synuclein pathology, are ideal for such neuroprotective trials.

Although nearly all patients with iRBD develop PD or DLB, not all people with PD or DLB develop iRBD first or go on to develop it at all. At motor symptom onset, only 30% of patients with PD have RBD. Similarly, only ~50% of patients with clinically manifest PD and 75% of people with DLB or MSA have RBD. However, these proportions might be underestimates of the occurrence of RBD over the entire course of the illness, as the frequency of polysomnography-verified RBD in late-stage PD and DLB has not been documented. In addition, dream enactment is a necessary criterion for the diagnosis of RBD but this symptom can disappear over time, possibly leading to underestimation of the proportion of individuals affected. Nevertheless, RBD is a key marker of diffuse-malignant PD subtypes. In this context, RBD is associated with cognitive loss, severe autonomic dysfunction, fast motor progression (particularly in gait) and loss of independent living and mortality¹². Similarly in patients with DLB, RBD is associated with specific clinical characteristics, including more parkinsonism,

other classic DLB hallmarks (hallucination, fluctuations) and a higher mortality¹⁰⁰.

Together, these findings imply that iRBD should be considered a specific subtype of prodromal PD or DLB. Cohort studies have shown that individuals with an RBD prodromal subtype usually develop bradykinesia and rigidity early and tremor evolves later, closer to the time of PD diagnosis⁷. By contrast, in non-RBD prodromal subtypes, tremor often precedes apparent bradykinesia and rigidity³³. This distinction in prodromal subtypes could be clinically important because if iRBD is a subtype of prodromal PD and DLB, insights gained from the study of patients with this condition might not apply to patients without it. For example, patients with PD but not RBD might be less likely to present with many of the distinctive non-motor features, such as autonomic dysfunction or detectable cognitive loss. Whether olfaction would differ between these prodromal subtypes is unclear. With fewer non-motor features, patients with non-RBD prodromal PD might be more difficult to identify. Moreover, given that these patients lack many of the longest-interval prodromal markers and are more likely to present with tremor, thereby resulting in an early diagnosis of PD, their observable prodromal state might also be considerably shorter than patients with an RBD prodrome, providing less opportunity to intervene with neuroprotective therapies.

Body-first and brain-first subtypes

The initial pathological aggregation and propagation of α -synuclein can occur in the brain or the periphery¹⁰¹, and this observation has led to the hypothesis that PD comprises two overall subtypes: a body-first (or bottom-up) subtype, in which α -synuclein pathology originates in the autonomic and enteric nervous system and invades the CNS via the vagus nerve and sympathetic connectome, and a brain-first (top-down) subtype, in which pathology arises in the brain itself, most often in the limbic system or sometimes secondary to entry via the olfactory bulb, and descends through the brainstem and into the periphery⁸¹ (FIG. 2). These subtypes align well with data from post-mortem studies and established neuropathological staging systems in PD and DLB. Post-mortem studies have identified two overarching profiles of Lewy pathology in the brain of patients with PD or DLB: a brainstem-predominant pattern of α -synuclein pathology that hypothetically corresponds with the body-first subtype, and a limbic-predominant pattern that corresponds with the brain-first subtype^{19,102–105}.

In this framework, premotor RBD is tightly linked to the body-first type, since ascending α -synuclein pathology affects pontine structures before the nigral dopamine neurons. By contrast, patients with the brain-first subtype usually do not have RBD when motor symptoms emerge, although top-down RBD can develop once the pons becomes damaged. Thus, body-first and brain-first PD are not synonymous with RBD-positive and RBD-negative PD. The relationship between RBD and brain-first and body-first subtypes of PD was investigated in a multimodal imaging study, which showed that patients with iRBD and

patients with de novo PD and premotor RBD follow the expected body-first trajectory, characterized by initial loss of cardiac sympathetic innervation and intestinal parasympathetic innervation followed by damage to the nigrostriatal dopamine system⁹⁶. By contrast, patients with de novo PD but no RBD follow the opposite trajectory: initial loss of the nigrostriatal dopamine innervation but relatively intact sympathetic and parasympathetic innervation⁹⁶ (FIG. 3).

If correct, the hypothesis has important implications for our understanding of Lewy body disorders. For example, some triggering factors, such as microbiome alterations^{106,107} and leaky gut syndrome, could predispose to the body-first subtype. In addition, the biomarkers that are relevant for prodromal diagnosis and monitoring of progression would be subtype-dependent. For instance, normal cardiac MIBG scintigraphy findings does not rule out prodromal PD, as most patients with the brain-first subtype initially have normal MIBG scans. Imaging-based strategies for monitoring disease progression would also be more complicated. Once identifiable at an early prodromal stage, progression of body-first PD might be best monitored with MIBG scintigraphy, as cardiac denervation commences 5–15 years before loss of the nigrostriatal dopamine system, whereas progression of prodromal brain-first PD would be best monitored with dopamine imaging, as loss of dopaminergic innervation begins years before the autonomic system is affected.

Important to keep in mind is that the body-first and brain-first subtypes should currently not be considered

a definitive classification of PD, but as clinicopathological phenotypes that explain a large degree of the phenotypic variance among patients with PD and reconcile clinical and imaging findings with longstanding neuropathological observations. However, other important factors, such as a genetic predisposition and variable selective neuronal vulnerability, are likely to underlie additional variance in clinical presentation and progression patterns¹⁰⁸. Furthermore, the brain-first and body-first hypothesis has been tested only recently in a single cohort of patients with PD⁹⁶. The findings must, therefore, be replicated by other researchers to determine their robustness. In particular, longitudinal studies in people with prodromal PD are needed to track the rate of neurodegeneration in each subtype, establish the rate of phenoconversion and determine to what degree prodromal brain-first and body-first subtypes can be unequivocally defined and what other variables modulate the phenotype and/or conversion.

Genetic subtypes

Genetics can determine clinical phenotypes of PD, so almost certainly have implications for prodromal phenotypes as well. Common genetic variants account for ~22% of the heritable risk of PD³⁰, and a high polygenic risk score for PD has been associated with more severe motor and cognitive decline in people with clinical PD¹⁰⁹. Whether low-penetrance genetic variants lead to phenotypic differences in the prodromal phase is still unclear. However, evidence suggests that genetic variants in single genes can define distinct prodromal phenotypes.

Retrospective studies suggest that people with mutations in *GBA* have a shorter but more severe prodromal phase than people with PD without these mutations¹¹⁰. *GBA* mutations are also more common among people with RBD than among the general population of people with PD. Among patients with iRBD, those with *GBA* mutations are indistinguishable from those with the general RBD prodromal phenotype, but their phenoconversion to clinical PD and DLB occurs faster¹¹¹. By contrast, *LRRK2* mutations, in particular the Gly2019Ser mutation, are associated with milder non-motor symptoms¹¹². Neither olfactory loss nor RBD are prominent features in prodromal *LRRK2* PD¹¹³.

These observations suggest that different prodromal phenotypes are inherent to genetic subgroups of *GBA* and *LRRK2* mutation carriers. Interestingly, on the basis of the clinical profiles associated with these mutations, *GBA* and *LRRK2* mutations might predispose to body-first and brain-first PD, respectively. Nearly all symptomatic patients with *GBA* mutations have abnormal MIBG scan findings, whereas most people with *LRRK2* mutations have normal MIBG scan findings⁸¹. The degree to which the predictive values of risk and prodromal markers in genetic subgroups are comparable with those in purely idiopathic forms of PD requires further investigation.

Biological subtypes

Biological subtypes of prodromal PD are an attractive concept, as such subtypes imply the possibility of developing targeted causal therapy. Biological subtypes would be defined on the basis that a specific biological

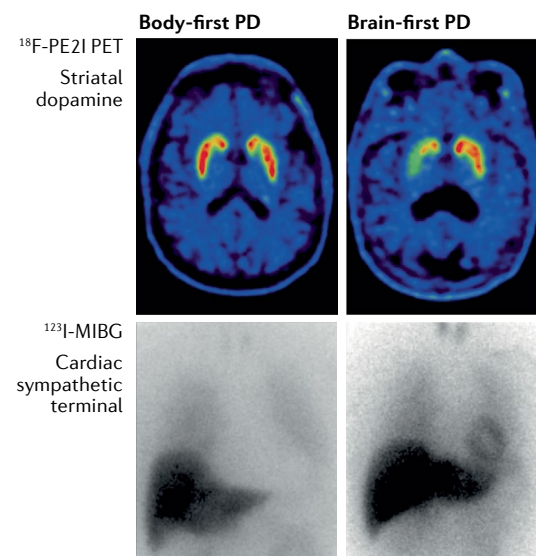


Fig. 3 | Imaging markers of body-first and brain-first PD subtypes. Striatal dopamine transporter imaging with ¹⁸F-PE2I PET and cardiac sympathetic terminal imaging with ¹²³I-MIBG scintigraphy shows that a patient with isolated REM sleep behaviour disorder (RBD; left) has normal dopaminergic innervation but severely depleted cardiac sympathetic innervation, which indicates body-first Parkinson disease (PD). By contrast, a patient at Hoehn and Yahr stage I of PD without RBD has marked unilateral dopaminergic denervation but normal cardiac sympathetic innervation, which indicates brain-first PD.

pathomechanism is the main cause of neurodegeneration and differentiates one group of individuals with prodromal PD from others. However, the evidence for specific biological subtypes even of manifest PD is currently limited.

Several biological pathomechanisms (such as mitochondrial dysfunction, increased tissue iron content, autophagy, calcium dysbalance and inflammation^{114,115}) could contribute to PD pathogenesis, but we cannot currently ascribe a unique subtype to each. These pathomechanisms might all contribute to many, if not all, subgroups of PD to different degrees. The degree to which a single pathomechanism contributes might be most pronounced in genetic forms of PD, but even in these forms, multiple downstream mechanisms can be involved. Therefore, further work needs to be done before biological subtypes can be implemented.

Pathophysiology and heterogeneity

Analysis of phenotypes to identify subtypes is only a starting point for identification of biological differences between different prodromal subtypes. For example, RBD is only a ‘wedge’ that enables body-first and brain-first phenotypes to be separated. This is not a perfect separation — exceptions exist — and alternative ‘wedges’ exist. Nevertheless, classification of subtypes on the basis of RBD led to a specific biological hypothesis that is testable and has clinical implications.

The heterogeneity of PD and its prodromal states indicates that the underlying mechanisms of prodromal states must differ. In this section, we consider some possible mechanisms that could underlie prodromal heterogeneity.

Variable spread of synucleinopathy

Aggregation of α -synuclein into Lewy bodies and the associated neurodegeneration can occur well beyond the substantia nigra during the prodromal stage of PD, suggesting that the pathological heterogeneity of PD is also present at the prodromal stage^{116–123}. Previous evidence of Lewy body pathology in individuals without clinical symptoms — defined as incidental Lewy body disease¹²⁴ — already suggested slow preclinical progression of pathology, and this process was conceptualized in the Braak staging scheme¹⁰², in which the dorsal nucleus of the vagus was proposed as the starting point of pathology (with a possible role in the onset of olfactory loss).

Entry of α -synuclein pathology into the brain from the gastrointestinal system might be particularly important in prodromal PD as suggested by the presence of gastric α -synuclein immunoreactive inclusions in the Meissner and Auerbach plexuses¹²⁵. A role for the vagal nerve in propagation of α -synuclein pathology from the gut to the brain^{126,127} was proposed by Hawkes et al.¹²⁸, and is supported by findings that vagotomy reduces the risk of PD in humans¹²⁹ and interrupts the spread of α -synuclein pathology in mice^{130,131}. The enteral microbiome could also contribute to inflammation and α -synuclein aggregation¹³². In a large study of the gut microbiome in healthy people, changes in the composition of the microbiome were associated

with several risk factors for and prodromal markers of PD, including physical inactivity, male sex, constipation, not smoking and subthreshold parkinsonism¹⁰⁶. However, the bidirectional links between these markers, microbial changes and the risk of PD need further investigation.

Nevertheless, the Braak staging scheme might not reflect the full heterogeneity of prodromal and clinical PD^{133,134}. Several studies have suggested that exceptions to the Braak spreading pattern occur in a substantial minority of patients^{135,136}, and that antero-grade and retrograde propagation of α -synuclein can occur via the vagus nerve^{137–139}; in this context, the gut–brain hypothesis might need to be reconsidered. Furthermore, the Braak staging model does not explain cases in which dementia (reflecting synuclein pathology) starts first¹⁴⁰.

Several alternatives to the Braak model have been proposed. A spectrum of Lewy body disease has been proposed that incorporates PD with and without dementia and DLB as three subtypes with different pathological spread¹⁴¹. In another model, pathology in the olfactory bulb is proposed as stage 1 in all Lewy body disease, followed by either brain-stem-dominant spread or limbic-predominant spread in stage 2 (REF.¹⁹).

The differences in pathological spread might simply be a reflection of the heterogeneity in Lewy body diseases, and as yet undetermined factors could contribute to the variability in pathological spreading and need further investigation. In particular, additional pathologies, such as amyloid- β and tau, can interact with Lewy pathology and could, therefore, influence progression routes and clinical subtypes^{103,142–144}. Such factors have not been incorporated into current staging schemes of Lewy pathology (see Non-PD pathologies). Furthermore, all staging schemes might fail to incorporate genetic forms of PD in which Lewy pathology is low or even absent¹⁴⁵.

Variable degenerative thresholds

Not all tissues are necessarily equally susceptible to damage from α -synuclein. According to a ‘threshold theory’ proposed in 2017 (REF.¹⁴⁶), parallel degeneration pathways might occur in the CNS and PNS but the functional threshold at which early symptoms emerge is reached sooner than that for the classic motor features¹⁴⁶. This hypothesis could explain why symptoms such as hyposmia, RBD, constipation and other dysautonomias manifest early in the prodromal stage of PD. In a similar theory of susceptibility, degeneration in the substantia nigra and other nuclei is thought to be accelerated, perhaps owing to high energy demands as a result of high baseline activity and arborization²⁴ or to certain types of pacemaker calcium channels²³.

These threshold theories are largely based on findings from animal models and do not account for the observed temporal evolution of damage to different neuronal systems or for the observed propagation of α -synuclein. Nevertheless, the underlying principle suggests another potential source of heterogeneity in PD — that is, patients might have different region-specific functional thresholds, meaning that different manifestations emerge first,

thereby leading to different prodromal stages. For example, individuals who first develop RBD and subsequently develop a diffuse-malignant clinical subtype might have a relatively resistant substantia nigra, resulting in later phenocconversion to parkinsonism despite advanced neurodegeneration elsewhere. By contrast, patients who present with a 'benign motor-dominant' subtype might have a relatively low threshold in the dopaminergic substantia nigra, leading to early parkinsonism and later development of non-motor symptoms such as RBD, dysautonomia and cognitive deficits.

Factors that modulate pathology

The idea that PD should be seen as an umbrella term that encompasses a spectrum of disease has been discussed above (see Clinical heterogeneity). The heterogeneity across this spectrum might be even more evident in the prodromal phase when the motor features that unite the heterogeneous forms have not developed. Moreover, neurodegenerative pathologies other than Lewy pathology — including amyloid- β and tau pathologies, vascular pathologies and neuroinflammatory pathology — could contribute to the heterogeneity.

Non-PD pathologies. Approximately 20% of people aged 65 years and >30% of people aged 80 years with normal cognition exhibit abnormal amyloid- β deposition. Amyloid- β deposits are known to accelerate α -synuclein pathogenesis¹⁴⁷, so co-existent amyloid- β pathology in PD and DLB could drive 'cortical-first' subtypes in which dementia presents early. Analysis of CSF has shown that markers of tau pathology are associated with a more severe motor progression, whereas markers of amyloid- β pathology are associated with future cognitive decline in PD¹⁴⁸. These findings suggest that coincident amyloid- β and tau pathologies could underlie subtypes of clinical and prodromal PD.

Other factors. Ageing is associated with several natural yet pathological alterations at the molecular, cellular, organ and system levels¹⁴ that could increase the selective vulnerability of certain neuronal groups and/or synergistically drive the progression of PD-specific pathologies. For instance, age-related malfunction of proteostasis mechanisms could contribute to α -synuclein misfolding¹⁴⁹, resulting in a diffuse rather than a 'mitochondrial' pattern of selective substantia nigra degeneration, as is seen in young-onset genetic PD.

Environmental factors could also contribute to prodromal and clinical subtypes. Such factors are usually thought of as risk factors, but causative environmental factors could modulate phenotype. For example, a mitochondrial toxin, such as a pesticide, could specifically target the substantia nigra and related brainstem nuclei, resulting in a pure motor phenotype in which non-motor manifestations and dementia occur late. In this scenario, few non-motor markers would be detectable in the prodromal phase. Alternatively, a protective environmental factor could slow degeneration in the substantia nigra so that a patient might present relatively late in the course of their synucleinopathy with a non-motor-predominant phenotype.

Genetic factors also modulate prodromal subtypes, as described above. Beyond direct effects of gene mutations on disease, genetics can also modulate the effects of environmental factors such as pesticide exposure¹⁵⁰, smoking¹⁵¹ and head injury¹⁵². Therefore, genetics might need to be considered for stratification of prodromal PD subgroups on the basis of some environmental risk factors. Finally, level of education is linked to differences in motor and cognitive reserve — highly educated people with PD have better baseline motor and cognitive function than less highly educated patients¹⁵³. Age of onset and progression did not depend on level of education in this study, but compensatory and reserve functions could be relevant to the prodrome of PD.

Conclusions and implications

On the basis of our current understanding, recognition of prodromal PD subtypes could influence the field in several areas. The first of these is in understanding disease mechanisms. Clinical symptoms of prodromal disease have a pathophysiological correlate, and an understanding of the variability in prodromal subtypes provides an opportunity to understand the mechanisms by which patients can differ. In turn, understanding these variable mechanisms could lead to treatments that can alter phenotype and prognosis.

The second area is the prediction of progression and phenocconversion. Within clinical PD, prognosis can differ substantially between subtypes, and identification of corresponding subtypes in early disease will provide prognostic information about the progression of neurodegeneration and the expected time of conversion to clinical PD. This improvement in prognosis has two major implications. First, accurate prediction of the time to conversion is an important basis for disease-modifying trials, which currently have no suitable outcome parameters. Second, accurate prediction of disease progression and manifestation of clinical symptoms is important for individuals with probable prodromal PD, who currently face high uncertainty in the absence of treatments.

A more complete understanding of prodromal subtypes will also enable us to determine the biases caused by diagnostic difficulties. As discussed above, relatively diffuse subtypes would become easier to diagnose on the basis of diagnostic markers for these subtypes, enabling recruitment of a broader spectrum of patients into trials of neuroprotective therapies.

The existence of prodromal PD subtypes also has implications for mechanism-based diagnostics. Specifically, work to identify and validate diagnostic biomarkers of prodromal PD should take into account the value of these biomarkers in different subtypes. As an example, changes in the microbiome, findings of gastrointestinal biopsies and imaging markers could have different values in the proposed body-first and brain-first subtypes. Finally, prodromal PD subtypes have similar implications for the development of mechanism-based therapies. In planning a trial of a neuroprotective therapy in prodromal PD, one must consider whether prodromal subtypes defined on the basis of phenotype alone correspond to specific biological disease mechanisms that could be targeted by the drug of interest.

Documented examples of a relationship between phenotype and pathological mechanisms in PD are scarce. However, these relationships are conceivable. For example, given the strong connection between iRBD, diffuse synucleinopathy and *GBA* mutations, people with iRBD might be good candidates for trials of therapies that target *GBA* or its product, glucocerebrosidase. By contrast, people with iRBD might not be suitable for trials of therapies that target *LRRK2*, as RBD is relatively uncommon among patients with *LRRK2* mutations. Similarly, trials of a therapy that targets mitochondrial function might require enrolment of patients with prodromal subtypes

with demonstrable mitochondrial dysfunction, such as those with *PRKN* or *PINK1* mutations.

These implications make clear that knowledge of subtypes of prodromal PD as well as clinical PD is crucial for addressing PD with a precision medicine approach. Knowledge of prodromal subtypes can be applied to the design of more efficient personalized or subtype-specific, multidisciplinary, patient-centred management approaches that can be initiated early in the prodromal stage.

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